The Fixed-Epsilon Renorming From arXiv:2301.07943 Is Not LASQ

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Source Problem

The source paper is Jose Rodriguez and Abraham Rueda Zoca, *On weakly almost square Banach spaces*, arXiv:2301.07943 [1]. On page 3 of the arXiv PDF, after recalling that slice-D2P and D2P are distinct, the authors ask whether there exists a locally almost square (LASQ) Banach space which fails D2P.

have been widely studied in the literature (see, e.g., [6, 7, 18]).

The techniques of Theorem 1.1 allow us to show the Lebesgue-Bochner space $L_1(\mu, Y)$ is WASQ for any Banach space Y whenever μ is a non-atomic finite measure (Corollary 2.5). This result should be compared with the above mentioned result of [27] that the property of being LASQ passes from a Banach space Y to the Köthe-Bochner space $E(Y)$, for any Banach function space E. We finish Section 2 with an example of a WASQ Banach space of the form $L_1(\nu)$ as in Theorem 1.1 which is not an $\mathcal{L}_1$-space (Subsection 2.3).

In Section 3 we go a bit further in the analysis of the link between almost squareness and diameter two properties. One of the main questions raised in [2] is whether there exists a LASQ Banach space which is not WASQ. Very recently, Kaasik and Veeorg proved in [29, Section 2] that the answer is negative and that an example can be found in the class of Lipschitz-free spaces over complete metric spaces. For such spaces, the properties SD2P, D2P, slice-D2P and LASQ are equivalent (combine [9, Theorem 1.5] and [24, Theorem 3.1]), so the above mentioned example satisfies the SD2P. Since the slice-D2P and the D2P are different properties [10], it is a natural question whether there exists a LASQ Banach space which fails the D2P. Within the framework of Banach lattices, a stronger version of the LASQ property which implies the D2P has been considered in [16]. Even though we do not know the answer to the previous question, we make some progress in this direction. Our main result in Section 3 is the following:

Their Theorem 1.2, shown below from page 4, proves a near-square renorming result for spaces containing a complemented copy of c_0.

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Theorem 1.2. *Let X be a Banach space containing a complemented isomorphic copy of c_0. Then for any $0 < \varepsilon < 1$ there exists an equivalent norm $|\cdot|$ on X such that:*

(i) *$(X, |\cdot|)$ has the slice-D2P, that is, every slice of $B_{(X,|\cdot|)}$ has diameter 2.*
(ii) *There are non-empty relatively weakly open subsets of $B_{(X,|\cdot|)}$ of arbitrarily small diameter.*
(iii) *$(X, |\cdot|)$ is (r, s)-SQ for all $0 < r, s < \frac{1-\varepsilon}{1+\varepsilon}$ in the sense of* [8, Section 6]*, that is, for every finite set $\{x_1, \ldots, x_n\} \subseteq S_X$ there exists $y \in S_X$ satisfying*

$$|rx_i \pm sy| \leq 1 \quad \text{for every } i \in \{1, \ldots, n\}.$$

Condition (iii) measures somehow how far is the norm from being ASQ. Notice that a Banach space is ASQ if and only if it is (r, s)-SQ for all $0 < r, s < 1$.

Theorem 1.2 applies to any separable Banach space containing an isomorphic copy of c_0, thanks to Sobczyk's theorem. The proof of Theorem 1.2 is inspired by the renorming technique developed by Becerra Guerrero, López-Perez and Rueda Zoca in [10, Theorem 2.4], which in turn uses ideas of the example of Argyros, Odell and Rosenthal [5] of a closed bounded convex subset of c_0 having the convex point

Result Type

This packet gives a **partial obstruction**. It proves that the concrete fixed-ε renorming used in the source proof of Theorem 1.2 is not LASQ. Thus the paper's construction cannot itself be the desired LASQ space failing D2P. The full LASQ-versus-D2P question remains open.

Proof Intuition

The source unit ball is built from three kinds of generators. The first two, $A \times \{0\}$ and $-A \times \{0\}$, have no second c_0-coordinate. The third kind can carry second-coordinate mass, but only after paying a fixed positive ε in the functional $L = (\lim, 0)$: those points have $L \geq -(1-\varepsilon)$, whereas the tree points $x_\alpha \in A$ have $L = -1$. Consequently, any point of the unit ball with L very close to -1 has very small second coordinate. At the particular point $(x_\emptyset, 0)$, the first coordinate is already coordinatewise on the boundary of the ambient sup-norm ball. If both $(x_\emptyset, 0) \pm y$ stay almost in the unit ball, then both the first and second coordinates of y must be small. This makes $(x_\emptyset, 0)$ strongly extreme, which is incompatible with LASQ.

Definitions And Source Construction

We use the notation of Section 3 of the source paper. Let

$$Z = c(\mathbb{N}^{<\omega}) \oplus_\infty c_0$$

with its usual norm $\|\cdot\|_Z$. For each $\alpha \in \mathbb{N}^{<\omega}$, define $x_\alpha \in c(\mathbb{N}^{<\omega})$ by

$$x_\alpha(\beta) = \begin{cases} 1, & \beta \preceq \alpha, \\ -1, & \text{otherwise.} \end{cases}$$

Put $A = \{x_\alpha : \alpha \in \mathbb{N}^{<\omega}\}$. The source fixed-$\varepsilon$ unit ball is

$$B = \overline{\text{conv}}\Big((A \times \{0\}) \cup (-A \times \{0\}) \cup \big((1-\varepsilon)B_Z + \varepsilon B_{c_0(\mathbb{N}^{<\omega})} \times \{0\}\big)\Big),$$

where $0 < \varepsilon < 1$. Let $|\cdot|$ be the Minkowski functional of B. The source proof notes that

$$(1-\varepsilon)B_Z \subseteq B \subseteq B_Z, \quad \text{hence} \quad \|z\|_Z \leq |z| \leq \frac{1}{1-\varepsilon} \|z\|_Z \,.$$

Let $P_2 : Z \to c_0$ be the second-coordinate projection and let

$$L(x, y) = \lim x.$$

A point $p \in S_{(Z,|\cdot|)}$ is strongly extreme if $|p \pm y_n| \to 1$ implies $|y_n| \to 0$.

Main Result

Lemma 1. *For every $w \in B$,*

$$\|P_2 w\|_\infty \le \frac{1-\varepsilon}{\varepsilon}\big(1 + L(w)\big).$$

Proof. Consider first the three generating pieces of B. If $w \in A \times \{0\}$ or $w \in -A \times \{0\}$, then $P_2 w = 0$, so the inequality is immediate. If

$$w = (1-\varepsilon)u + \varepsilon(v, 0)$$

with $u \in B_Z$ and $v \in B_{c_0(\mathbb{N}^{<\omega})}$, then

$$\|P_2 w\|_\infty \le 1 - \varepsilon$$

and

$$L(w) = (1-\varepsilon)L(u) \ge -(1-\varepsilon).$$

Therefore $1 + L(w) \ge \varepsilon$, and the desired inequality again holds.

The set of $w \in Z$ satisfying the displayed inequality is convex: the left side is convex and the right side is affine. It is also norm closed. Since it contains all generators, it contains their closed convex hull B. $\square$

Theorem 1. *For every $0 < \varepsilon < 1$, the fixed-ε renormed space $(Z, |\cdot|)$ constructed in the source proof is not LASQ. In fact, $p = (x_\emptyset, 0)$ is a strongly extreme point of B.*

Proof. First note that $p \in S_{(Z,|\cdot|)}$: $p \in A \times \{0\} \subseteq B$, while $B \subseteq B_Z$ and $\|p\|_Z = 1$.

Let $y = (u, v) \in Z$, and suppose that

$$|p + y| \le 1 + \delta, \qquad |p - y| \le 1 + \delta$$

for some $\delta \ge 0$. Since $B \subseteq B_Z$, we have

$$\|x_\emptyset \pm u\|_\infty \le 1 + \delta.$$

Every coordinate of $x_\emptyset$ has absolute value 1. Hence $|u(\beta)| \le \delta$ for every $\beta \in \mathbb{N}^{<\omega}$, so

$$\|u\|_\infty \le \delta \quad \text{and} \quad |\lim u| \le \delta.$$

Set $q_+ = (p + y)/(1 + \delta) \in B$. Lemma 1 gives

$$\frac{\|v\|_\infty}{1+\delta} = \|P_2 q_+\|_\infty \le \frac{1-\varepsilon}{\varepsilon}\left(1 + \frac{-1 + \lim u}{1+\delta}\right) = \frac{1-\varepsilon}{\varepsilon}\,\frac{\delta + \lim u}{1+\delta} \le \frac{1-\varepsilon}{\varepsilon}\,\frac{2\delta}{1+\delta}.$$

Thus

$$\|v\|_\infty \le \frac{2(1-\varepsilon)}{\varepsilon}\delta.$$

Combining the two coordinate estimates,

$$\|y\|_Z \le \max\left\{\delta, \frac{2(1-\varepsilon)}{\varepsilon}\delta\right\}.$$

Using the source norm comparison,

$$|y| \le \frac{1}{1-\varepsilon}\, \|y\|_Z \le \max\left\{\frac{1}{1-\varepsilon}, \frac{2}{\varepsilon}\right\}\delta.$$

Now let $y_n \in Z$ be any sequence such that $|p+y_n| \to 1$ and $|p-y_n| \to 1$. Applying the estimate with

$$\delta_n = \max\{|p+y_n|\,,|p-y_n|\} - 1$$

gives $|y_n| \to 0$. Therefore p is strongly extreme.

An LASQ space cannot have this behavior at a sphere point: LASQ would require a sequence $y_n \in B$ with $|p \pm y_n| \to 1$ and $|y_n| \to 1$. Hence $(Z, |\cdot|)$ is not LASQ. □

Corollary 1. *The fixed-ε construction in the proof of Theorem 1.2 of [1] cannot by itself provide a locally almost square Banach space failing D2P.*

Proof. The theorem shows that the core c_0-renorming used in the construction is not LASQ for any fixed $0 < \varepsilon < 1$. □

Verification Notes

No numerical computation is used. The only estimate requiring review is Lemma 1. It is a generator-by-generator convexity estimate: second-coordinate mass can occur only in the third generating piece, and that piece raises the $L = (\lim, 0)$ value by at least ε from the tree value -1.

Limitations And Novelty Check

This packet does not solve the full question whether there exists a LASQ Banach space failing D2P. It rules out only the most direct attempt to upgrade the source fixed-ε renorming to LASQ. It also does not rule out renormings with a variable parameter, different tree balls, or non-c_0 constructions.

Before packaging, the run's lightweight indexes were searched for arXiv:2301.07943, the paper title, weakly almost square, locally almost square, LASQ, WASQ, D2P, and diameter two phrases. No duplicate packet for this paper or this obstruction was found. A bounded web search on June 15, 2026 for exact phrases including "locally almost square" "diameter two property", "LASQ" "D2P", and close variants found the source paper and nearby LASQ lattice/transfinite-ASQ papers, but no later explicit answer to the full LASQ-fails-D2P question.

Human Review Recommendation

The proof is short and appears likely valid. Reviewers should verify that the closed convex hull argument in Lemma 1 is legitimate and that the coordinatewise boundary argument for $x_\emptyset$ indeed forces $\|u\|_\infty \le \delta$ whenever $p \pm (u, v) \in (1 + \delta)B$.

References

[1] J. Rodriguez and A. Rueda Zoca, *On weakly almost square Banach spaces*, arXiv:2301.07943, 2023.

[2] T. A. Abrahamsen, J. Langemets, and V. Lima, *Almost square Banach spaces*, Journal of Mathematical Analysis and Applications 434 (2016), 1549–1565.